

LANE A under C6 — the transition-map sweep: what closes, what closes-as-projection, what doesn't

C6 (Cal §654): report the transitions that don't close alongside the ones that do — the full sweep, not the hits. A sweep with honest nulls is more persuasive than a handful of matches. Headline of the bar (a method, not a check): the address is named before the value. Every row below names its (object, verb) pair first. Reconnected/pinned first: the 2026-08-12 genus relabeling (Bergman genus = $n_C = 5$, Hua 1963), T1918/Toy 2351 $((n+1)/n)$, the discrete-series weight $k = n_C + 1 = 6$, the $\lambda_2 > 0$ zero-Šilov-value (confinement engine), Harish-Chandra Plancherel.

The convention-pin (Bergman $\leftrightarrow$ Szegő) — 6/5 promotes, but as the WEIGHT ratio, not the singularity exponent

Pinned to Hua (1963) / the 2026-08-12 relabeling — two *different* well-defined exponents: - Bergman genus (kernel *singularity* exponent) = $n_C = 5$ (= dual-Coxeter $h^{\vee}(B_3)$); Szegő singularity exponent = $n_C - 1 = 4$. Singularity-exponent ratio = $5/4$. - Bergman-space discrete-series weight $k_B = n_C + 1 = 6$ ($A^2(D_{IV}^5)$ = holomorphic discrete series π_6 , Harish-Chandra); Hardy/Szegő weight $k_S = n_C = 5$. Weight ratio = $(n_C + 1)/n_C = 6/5$.

$\Rightarrow$ the banked “6/5 = Bergman/Szegő kernel exponent ratio” (T1918) is the *weight* ratio, mislabeled “kernel exponent.” Corrected (superseded-not-deleted): 6/5 is the discrete-series WEIGHT ratio $k_B/k_S = (n_C + 1)/n_C$; the *singularity-exponent* ratio is separately $5/4$ (genus $n_C=5$ / $n_C-1=4$). Both stay CANDIDATE — the numbers wait for Elie's read of the printed Hua (1963) page, not this corpus-derived weight argument (reconnect to the *primary source* before promoting; the corpus label was itself wrong, so the corpus is not the authority here). Write each ratio as its labeled pair — $(n_C + 1)/n_C$ for the weight, $n_C/(n_C - 1)$ for the singularity — never “ C_2/n_C ” ($C_2 = 6$ and $n_C + 1 = 6$ agree only at $n = 5$; same-number trap). Target-innocent (functions of n_C), disk control passed. (*Convention-before-value: the number may be right, but it promotes on the printed page, not on a corpus that mislabeled it.*)

The C6 sweep — every candidate transition among the four decompositions, tiered

transition (chart $\leftrightarrow$ chart)	verb / map	closes?	tier	induced reading (address $\rightarrow$ value)
Bergman(bulk) $\leftrightarrow$ Szegő(boundary)	Π (Szegő projection) / E (boundary- value ext.)	YES ($\Pi \circ E = \text{id}_{\{H^2\}}$, E injective by T349)	map DERIVED; weight-ratio & singularity- ratio CANDIDATE (pending printed Hua) DERIVED	(K_B, K_S) $\rightarrow$ weight ratio (n_C+1)/ n_C $= 6/5$, singularity ratio $n_C/(n_C-1)$ $= 5/4$ (K -type λ) $\rightarrow$ its radial eigenvalue on $A = C_2(\lambda)$ (Casimir $\leftrightarrow$ radial- Laplacian eigenvalue) a K -type reaches the Šilov boundary iff $\lambda_2 = 0$; K -types with $\lambda_2 > 0$ have zero Šilov value (the confinement / $m_1=0$ engine) — the kernel is the confined sector
Cartan slice $\leftrightarrow$ K -type (KAK)	Harish- Chandra (τ -spherical) transform	YES (Plancherel isomorphism)		
K -type $\leftrightarrow$ Šilov stratum (Korányi- Wolf)	boundary restriction	closes as a PROJEC- TION with a kernel	DERIVED- structure	

transition (chart $\leftrightarrow$ chart)	verb / map	closes?	tier	induced reading (address $\rightarrow$ value)
bulk $\leftrightarrow$ Šilov stratum (direct)	—	NO — does not close directly	(null)	must factor through the boundary- value map (Bergman $\leftrightarrow$ Szegő); there is no direct bulk $\rightarrow$ stratum transition that isn't the composite. Reported as a null.
Peirce block $\leftrightarrow$ Cartan slice	—	NO clean map — different de- compositions	(null)	the Jordan frame (c_1, c_2, V_{12}) and the KAK flat A are not related by a proved transition on an overlap; a naive identification fails the property test (they coincide only at the base point). Reported as a null.

Two of five close, one closes-as-projection, two are honest nulls. The nulls are the point of C6: bulk $\leftrightarrow$ stratum only closes as the Bergman $\leftrightarrow$ Szegő *composite* (not a new map), and Peirce $\leftrightarrow$ Cartan has no proved transition (I tested the property — a shared base point is not a chart overlap — not mere existence; the new rule).

The trace-normalization soft joint (Cal’s flag) — why trace-normalization, given no unification

The answer is separability: the hypercharge normalization is a trace over $\text{End}_K = \mathbb{C} \oplus \mathbb{H} \oplus \mathbb{R}$ that factorizes over the census blocks, and the $\mathbb{R}$ color block contributes zero ($U(\mathbb{R})$ is 0-dimensional). So the trace is effectively over the electroweak part $\mathbb{C} \oplus \mathbb{H}$ alone — it does not mix the color block in, hence needs no unification to be well-defined. Trace-normalization is right *precisely because* the census is block-diagonal and color is separable: you normalize hypercharge by a trace that never sees color. “It’s what NCG does” is replaced by “the census trace factorizes and color drops out” — a reason internal to BST’s own structure, consistent with the no-unification headline.

Tiers / C6 status

- Bergman $\leftrightarrow$ Szegő: map Derived; the two ratios $((n_C+1)/n_C = 6/5$ weight, $n_C/(n_C-1) = 5/4$ singularity) stay Candidate pending printed Hua (label corrected from “kernel exponent”; never “ C_2/n_C ”).
- Cartan $\leftrightarrow$ K-type: Derived (Casimir $\leftrightarrow$ radial eigenvalue).
- K-type $\leftrightarrow$ stratum: Derived-structure (projection; kernel = $\lambda_2 > 0$ confined sector).
- bulk $\leftrightarrow$ stratum, Peirce $\leftrightarrow$ Cartan: honest nulls, reported (C6).

Handoffs

- @Elie: confirm the Szegő singularity exponent = $n_C-1 = 4$ against Hua’s Cauchy–Szegő formula (settles the $5/4$); confirm the Hardy weight $k_S = n_C = 5$. On confirmation the two ratios ($6/5$ weight, $5/4$ singularity) both register cleanly.
- @Keeper: gate the label correction (T1918’s “ $6/5 = \text{kernel exponent ratio}$ ” $\rightarrow$ “weight ratio”; superseded-not-deleted) and the sweep tiers; ratify the two nulls as reported nulls, not gaps.
- @Grace: add the three closing/projecting transitions + the two nulls to the atlas table (C6 — nulls in the same table as the hits).

Lyra + Elie, 2026-08-21 (Lane A, C6 sweep). Convention DIAGNOSED (not yet pinned to the printed page): $6/5$ is the discrete-series WEIGHT ratio ($k_B = n_C+1=6$ / $k_S = n_C=5$), the singularity-exponent ratio is separately $5/4$ (genus $n_C=5$ / 4) — BOTH stay CANDIDATE pending Elie’s read of printed Hua (1963); the “kernel exponent ratio” label (T1918) corrected. Write each as its labeled pair, never “ C_2/n_C ” (same-number trap, agree only at $n=5$). Sweep of all five candidate transitions among the four decompositions: Bergman $\leftrightarrow$ Szegő and Cartan $\leftrightarrow$ K-type CLOSE (Derived); K-type $\leftrightarrow$ stratum closes as a PROJECTION with kernel = the $\lambda_2 > 0$ confined sector; bulk $\leftrightarrow$ stratum and Peirce $\leftrightarrow$ Cartan are HONEST NULLS (reported, C6 — tested the property, not existence). Trace-normalization answered by separability: the census trace factorizes, color ($\mathbb{R}$) drops out, no

unification needed. Address named before value throughout. Nothing pushed; CP
existence-only.